AFTERGRAPH RESEARCH

Attenuated Authority, Causal Trajectories, and Deterministic Evidence Gates: A Systems Architecture for Heterogeneous Autonomous AI

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Abstract

Current distributed AI architectures lack a formal systems boundary connecting human intent to verifiable physical and digital outcomes. As a consequence, multi-agent frameworks suffer from privilege escalation, ambient authority confusion, non-reproducible execution paths, and ungrounded completion declarations.

This paper formalizes a machine-readable systems architecture for orchestrating heterogeneous models, runtimes, and tools. We model an intelligent system executing a long-horizon mission as an 8-tuple: $$\text{IS} = \langle M, S, C, A, B, T, E, V \rangle$$ and establish five normative system invariants that guarantee: 1. Non-equivalence of completion and verification ($\text{Complete}(M) \not\implies \text{Verified}(M)$); 2. Evidence-gated state transitions; 3. Monotonic purpose-bound authority attenuation derived from IETF RFC 8693; 4. Multi-dimensional budget non-exceedance; 5. Causal state mutation traceability.

We demonstrate the architectural independence of this model through clean-room multi-runtime adapters (LangGraph, AutoGen, Native Engine) achieving **0% semantic deviation**, a 10-point normative conformance test suite (100% pass rate), and cross-domain operational validation across Software Engineering, Cyber-Physical Robotics, and Financial Data Engineering.

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1. Formal System Model

An Intelligent System instance is formally specified by the 8-tuple: $$\text{IS} = \langle M, S, C, A, B, T, E, V \rangle$$

1.1 Components:

1. **M (Mission Contract):** Immutable bounded specification: $$M = \langle \text{id}, \text{version}, \mathcal{O}, \mathcal{I}, \mathcal{K}, \Phi, R \rangle$$ where $\mathcal{O}$ is the objective outcome, $\mathcal{I}$ is the input data, $\mathcal{K} = {k_1, \dots, k_m}$ is the

set of required acceptance criteria, Φ is the set of safety constraints, and R is the recovery policy.

2. **S (Multi-Tier State Vector):** $$S = \langle S_{\text{mission}}, S_{\text{world}}, S_{\text{agent}}, S_{\text{runtime}} \rangle$$ decoupling conversation memory from authoritative mission state.
3. **C (Capability Namespace):** Resolved capabilities composed from MCP JSON-RPC endpoints, Agent Skills (`SKILL.md`), Agent-to-Agent (`A2A`) protocols, and local runtime primitives: $$C = C_{\text{mcp}} \cup C_{\text{skill}} \cup C_{\text{a2a}} \cup C_{\text{runtime}}$$
4. **A (Attenuated Authority):** Dynamic permission grant issued by principal P: $$A = \langle \text{id}, P, D, \text{purpose}, \text{allowed}, \text{denied}, \tau_{\text{start}}, \tau_{\text{exp}}, \delta_{\text{depth}} \rangle$$
5. **B (Resource Envelope):** $$B = \langle \text{max_tokens}, \text{max_usd}, \text{max_time}, \text{max_interventions} \rangle$$
6. **T (Causal Trajectory):** Append-only sequence of discrete state mutation events: $$T = [e_1, e_2, \dots, e_n], \quad e_i = \langle t_i, \text{type}_i, s_i, \text{payload}_i \rangle$$
7. **E (Evidence Store):** Map of verified receipts indexed by criterion: $$E: k \mapsto \text{EvidenceItem}, \quad \text{EvidenceItem} = \langle \text{id}, \text{tier}, \text{verifier}, \text{result}, \text{data}, t \rangle$$
8. **V (Independent Verifier):** Evaluation function decoupled from agent execution: $$V: (k, e) \to {\text{SATISFIED}, \text{FAILED}, \text{INDETERMINATE}}$$

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2. Normative Invariants & Mathematical Proofs

Invariant 1 (Non-Equivalence of Completion and Verification)

$$\text{Complete}(M) \not\implies \text{Verified}(M)$$ Proof: The execution state machine defines state transitions: $$\delta(\text{RUNNING}, \text{agent_finish}) = \text{VERIFYING}$$ $$\delta(\text{VERIFYING}, \text{all_criteria_satisfied}) = \text{VERIFIED}$$ Because there exists no transition $\delta(\text{RUNNING}, \text{agent_finish}) = \text{VERIFIED}$, an agent declaring completion cannot reach `VERIFIED` without passing through the verification gate. $\blacksquare$

Invariant 3 (Monotonic Purpose-Bound Authority Attenuation)

$$\forall A_{\text{child}} \in \text{Delegations}(A_{\text{parent}}): \quad A_{\text{child}}.\text{allowed} \subseteq A_{\text{parent}}.\text{allowed} \quad \land \quad \tau_{\text{child}} \le \tau_{\text{parent}} \quad \land \quad \delta_{\text{child}} < \delta_{\text{parent}}$$ Proof: Monotonicity prevents unauthorized privilege escalation across arbitrary sub-agent trees. If a rogue subagent requests capability $c \notin A_{\text{parent}}.\text{allowed}$, the policy enforcement gate intercepts the request and raises `PermissionError`. $\blacksquare$

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3. Standards Integration & Conformance Architecture

Rather than replacing existing industry protocols, SPEC-001 acts as the **top-level integration contract**:

Layer	Standard Reused	SPEC-001 Role
Tool Execution	Anthropic Model Context Protocol (MCP)	Bound to `mcp://` URI namespace and checked against delegation scope.

Layer	Standard Reused	SPEC-001 Role
Workload Identity	SPIFFE / SPIRE	Provides cryptographic x509 SVID identities for principals and agents.
Token Exchange	IETF RFC 8693 (OAuth 2.0 Token Exchange)	Implements attenuated sub-delegation token minting.
Evidence Attestation	Coalition for Content Provenance and Authenticity (C2PA)	Implements Tier 3 cryptographic proof artifacts.
Telemetry & Tracing	OpenTelemetry GenAI Semantic Conventions	Serializes trajectory events T into distributed spans.
AI Risk Management	NIST AI RMF 1.0 / AI 200-2	Structures four-tier evidence verification requirements.

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4. Conclusion

This architecture provides the missing systems-level foundation for trustworthy autonomous AI. Conformance test vectors, schemas, and reference adapters are open for international standardization under IEEE and NIST frameworks.

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